CMDNet: A Cross-Modality Spatiotemporal Graph Network for Enhanced Air Pollution Prediction With High-Resolution Satellite Data

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***Abstract*— Predicting air pollution plays a vital role in urban management and public health by providing early warnings on PM2.5, SO_2, and NO_2 concentrations, helping to mitigate the adverse effects of these pollutants. Traditional prediction methods, relying on physical and statistical models, often struggle to capture the complex spatiotemporal dependencies and dynamic characteristics of air pollution data. The application of deep learning methods, especially graph neural networks (GNNs), has shown promise in addressing these limitations. However, existing GNN-based methods ignore the integration of rich semantic information provided by high-resolution satellite data. To address this problem, we propose a cross-modality dynamic spatiotemporal graph neural network (CMDNet) for air pollution prediction. The model comprises two branches: a dynamic spatiotemporal GNN (DSTGNN) branch and a remote sensing image dynamic encoding network (RSIDEN) branch. The DSTGNN branch captures the spatiotemporal dependencies in air pollution data by constructing a dynamic graph structure. The RSIDEN branch extracts semantic information from high-resolution satellite data, which improves the model's power to perceive air pollution conditions in different regions. Experiments on real-world datasets demonstrate that the CMDNet achieves better air pollution prediction results than existing SOTA models, with maximum improvements of 2.4% (MAE), 1.8% (RMSE), 1.3% (CSI), 1.6% (FAR), and 1.4% (POD), providing more accurate prediction results.**

***Index Terms*— Air pollution prediction, graph neural network (GNN), high-resolution satellite data, remote sensing image encoding.**

I. Introduction

AIR pollution has posed a significant threat to human health, ecosystem functionality, and climate change [1]. Air pollutants contain particulate matter such as PM2.5 and PM10, SO_2, CO, and nitrogen oxides such as NO_2. These pollutants lead to increased incidence of respiratory and cardiovascular diseases [2]. Large amounts of pollutants also cause phenomena such as acid rain and smog, exacerbating the greenhouse effect [3]. Consequently, air pollution forecasting has become a critical area of research. Accurate air pollution forecasts can inform the formulation of effective environmental policies [4]. With the advancements in remote sensing technology, monitoring data related to air pollution has been continuously collected, facilitating more precise predictions. Traditional methods in air pollution prediction research primarily involve physical modeling and statistical approaches. Physical models [5], [6] simulate the spread and diffusion of air pollutants using a series of physical equations that require detailed environmental parameters and initial conditions, such as meteorological conditions and topography. However, these models are sensitive to data errors, which increase the uncertainty of predictions. Additionally, the computational complexity of physical models escalates when handling large-scale data. Meanwhile, models based on statistical methods such as ARIMA [7] and Kalman filter [8] have been widely used for air pollution prediction. While these methods can somewhat forecast trends in air pollution, they have limitations in addressing complex spatiotemporal dependencies. These models typically assume that data are linear and independently distributed. In contrast, actual air pollution data often exhibit intricate nonlinear and dynamic characteristics that are difficult to capture accurately with simple linear models. Consequently, traditional methods exhibit limitations in both flexibility and accuracy in predicting air pollution.

In recent years, the development of deep learning methods has brought new tools to the task of air pollution prediction. Models such as convolutional neural networks (CNNs) [9] and long short-term memory networks (LSTMs) [10] have been used to learn the characteristics of air pollution data, significantly enhancing prediction accuracy. However, the above models typically concentrate on sequence features and may have difficulty in fully representing the dynamic correlations among air pollution data. Graph neural networks (GNNs) effectively capture the correlations of air pollution conditions across different regions by modeling spatiotemporal dependencies within a graph structure and utilizing a

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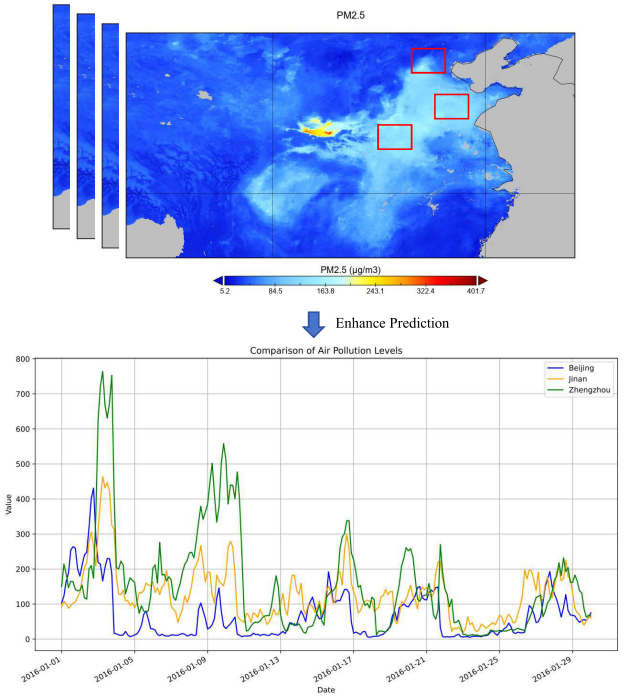


Fig. 1. Upper part of the figure shows remote sensing data, while the lower part displays the air pollution time-series data for three cities. The three boxes in the upper part correspond to the surrounding areas of the cities depicted in the lower part. In top-to-bottom order, the three boxes represent Beijing, Jinan, and Zhengzhou, respectively. The remote sensing images provide additional information about pollutant distribution in the areas surrounding each city, which complements the time-series data in the lower part of the figure by offering spatial context for the pollution levels observed over time.

message-passing mechanism between nodes. Based on GNNs, a series of air pollution prediction models have been proposed, including the GAGNN [11], SSH-GNN [12], and MST-GCN [13]. Although existing models based on monitoring station data have achieved good predictive performance, they overlook the additional information provided by remote sensing images, which still limits the accuracy of the predictions. High-resolution satellite data hold significant importance in tasks like environmental monitoring and air pollution prediction. It provides large-scale, continuous semantic information, including surface temperature and aerosol optical thickness. This semantic information can help prediction models learn the patterns and trends of air pollution and its spread. For example, the aerosol optical thickness is related to the concentration of particulate matter in the air, which can reflect the degree of air pollution. By analyzing remote sensing data related to air pollution, prediction models can further capture the spatial dependencies and variation patterns of pollutant concentrations [14].

As shown in Fig. 1, the upper part of the figure displays satellite remote sensing data, while the lower part shows air pollution monitoring sequences obtained from ground stations. Most existing prediction models rely on ground-based monitoring data. While these data offer high temporal resolution, their spatial coverage is limited, making it challenging to comprehensively reflect air quality conditions on a large scale. Additionally, ground monitoring data is often limited by the distribution of monitoring stations, making it unable to provide continuous air pollution information. This limitation restricts the model's ability to achieve more accurate prediction results. Fig. 2 illustrates air pollution in certain regions of China across different seasons, using PM2.5 as an example. It can be observed that remote sensing data offers a more global and comprehensive view of pollution patterns on different dates. Compared to monitoring sequence data, remote sensing data covers a larger spatial range and contains richer semantic information related to urban pollution, such as the distribution patterns of pollution around cities and the characteristics of pollution spread. Existing models mainly focus on modeling single-type data and lack joint learning of dynamic features from two modalities. Deep learning models such as GNNs can effectively predict monitoring sequence data, but methods that fully utilize remote sensing data for air pollution prediction remain relatively scarce. MF-net incorporates a multimodal fusion network that employs 3-D remote sensing data to predict pollutants [15]. STI-GCN introduces very high-resolution image features to estimate PM2.5 levels [16]. Unlike existing models, CMDNet integrates high-resolution satellite data and ground monitoring data to extract features from multimodal air pollution data. By jointly learning spatiotemporal dependencies from both ground-based data and remote sensing images, the CMDNet captures a more comprehensive view of air pollution patterns than previous models.

To use remote sensing data to improve air pollution prediction results, this study introduces a cross-modal dynamic spatiotemporal GNN (DSTGNN), which consists of a DSTGNN branch and a remote sensing image dynamic encoding network (RSIDEN) branch. The DSTGNN branch focuses on learning the dynamic spatiotemporal dependencies of air pollution data. It constructs a dynamic graph structure to capture correlations between different time steps and designs spatiotemporal graph networks to learn the spatiotemporal representation of air pollution data, effectively addressing the dynamic variations and spatial correlations within the data. Remote sensing data can help models to reveal the diffusion pathways and propagation mechanisms of pollutants in the atmosphere. The RSIDEN branch employs a Visual Transformer to learn semantic features related to air pollution from high-resolution satellite data, enhancing the model's ability to perceive air pollution conditions in different regions.

The main contributions of this study include the following.

1) The dynamic graph learning layer is proposed to capture the spatiotemporal dependencies of air pollutants, considering their spatial diffusion and a data-driven mechanism is used to model remote sensing data over time, thus improving the accuracy of pollutant transmission modeling.
2) Ground station and satellite remote sensing data are integrated to leverage their complementary spatial and temporal coverage. The physical correlations of these data sources are modeled to enhance the accuracy of air pollutant concentration predictions.
3) A novel cross-modal dataset, combining ground and high-resolution satellite data, has been constructed. The experimental results on this dataset indicate that the

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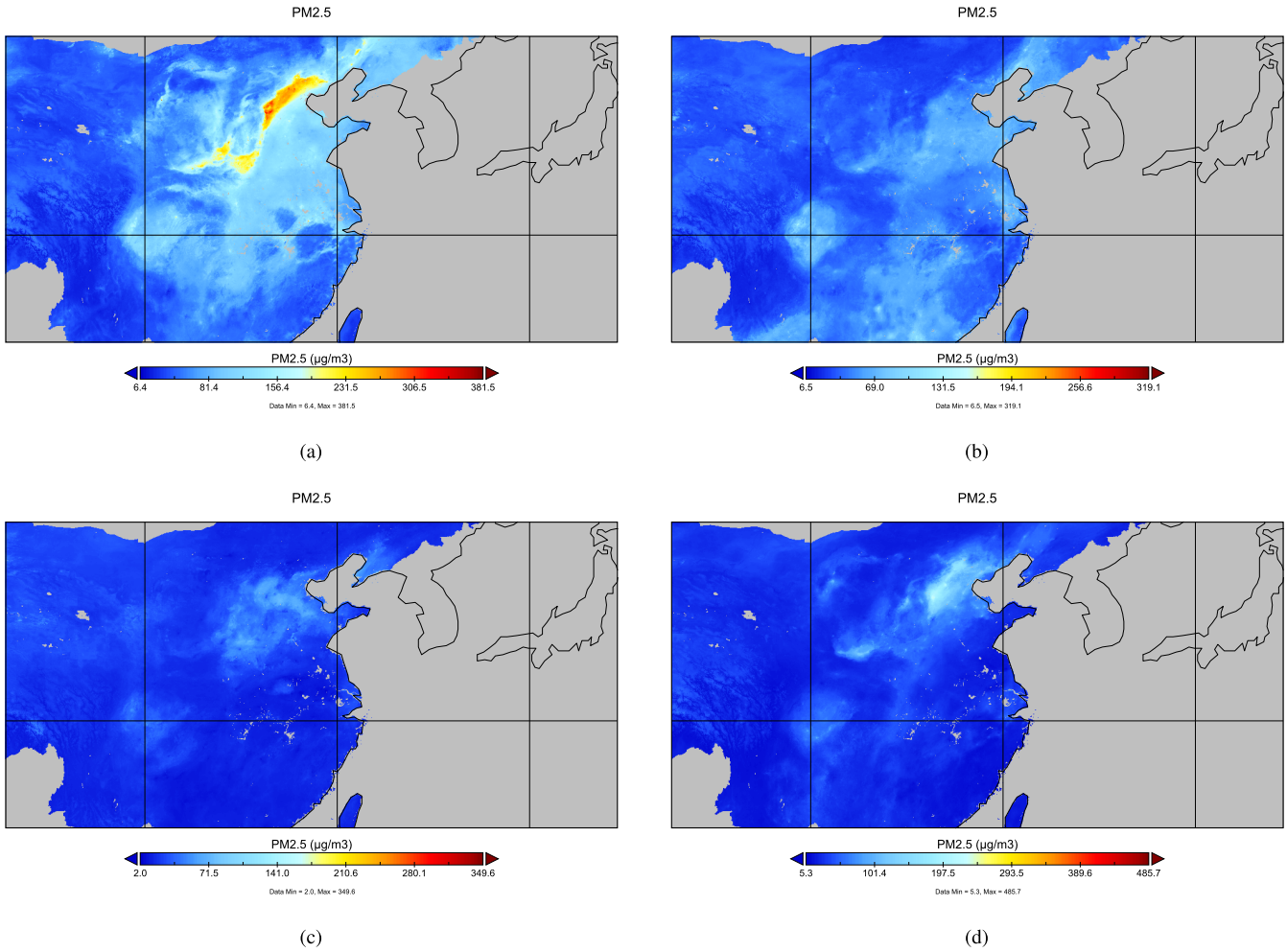


Fig. 2. Air pollution in different seasons across various regions of China, taking PM2.5 as an example. Subplots (a)–(d) represent satellite remote sensing images of PM2.5 pollution on the first day of four different months in 2016. The variations in air pollution patterns across the seasons reflect different pollution dispersion modes. Dynamic modeling of air pollution using remote sensing data is of great significance for accurate prediction, as it captures seasonal changes and their effects on pollution distribution. (a) January 01, 2016. (b) April 01, 2016. (c) July 01, 2016. (d) October 01, 2016.

CMDNet model outperforms the existing baseline models in the air pollution prediction task.

II. Related Works

Our study highlights the background literature in three main categories which consist of previous methods for air pollution predictions using traditional and deep learning techniques, GNNs applied to spatiotemporal modeling and remote sensing. All these aspects add diverse knowledge and methods that assist us in devising CMDNet for enhanced air pollution forecasting.

A. Air Pollution Prediction Methods

1) Traditional Models: Due to the serious problems that air pollutants have brought to society and the environment, air pollution prediction has become a key focus for scientific researchers. Consequently, these predictions must be made more accurately to minimize their impacts. Traditionally, air pollution prediction methods have primarily been based on ground monitoring data. These methods usually apply the corresponding statistical and machine learning methodologies [17], [18], like ARIMA [19], time-series analysis [20], support vector machines (SVMs) [21], and random forests [22]. The strengths of data coming from ground-based monitoring are mainly in the high temporal resolution of such information, which is real-time; however, in traditional methodologies of air pollution predictions, the capture of complex spatiotemporal dynamics of air pollution remains a substantial challenge. Several factors, including meteorological conditions, geographical environments, and human activities, govern the generation and dispersion of air pollutants, resulting in highly complex and dynamic spatiotemporal distributions. Given the very complex conditions with most existing models, it is tough to grasp the dynamic changes and their dispersion patterns for the prediction to be accurate and reliable.

2) Deep Learning Models: In the past decade, deep learning technology has rapidly developed. Deep learning models have shown great effectiveness in handling high-dimensional and nonlinear data, making them particularly suitable for the

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air pollution prediction task. For example, C-LSTME [10] constructs a convolutional LSTM network to predict air quality values. Utilizing domain knowledge about air pollution, DeepAir [23] comprises a spatial feature transformation module and a deep distribution fusion network to achieve more accurate prediction results. Airformer [24] employs a self-attention mechanism to model the spatiotemporal features, achieving accurate air prediction results. Many studies combine GNNs and temporal networks to predict air pollutant values [12], [25], [26], [27]. Furthermore, some studies [28], [29] combine multimodal data, such as meteorological and traffic data, to improve prediction robustness and accuracy.

Existing prediction models based on deep learning methods are primarily constructed with the aim of mining and analyzing ground monitoring data. However, due to the sparse spatial distribution of these stations and the complex spatial and temporal characteristics of air pollution, these methods have many shortcomings in terms of spatial coverage and prediction accuracy. Deep learning methods offer new possibilities for forecasting air pollution due to their ability to automatically extract complex features. The CMDNet model aims to address these limitations in traditional approaches by combining remote sensing data and dynamic GNNs.

3) Remote Sensing for Air Pollution Monitoring: The data obtained from remote sensing satellites has brought new research perspectives to the issues of air pollution monitoring or prediction. Large-scale continuous spatial information by remote sensing technology brings unique advantages to air pollution monitoring. Satellite sensors can measure many atmospheric attributes, including but not limited to aerosol optical thickness, surface temperature, and vegetation cover, which provides rich information for air pollution assessment. For example, MODIS and Landsat satellite high-resolution image data have been used widely to estimate PM2.5 concentrations in the atmosphere [30], [31]. These studies can demonstrate that the information obtained through the analysis and mining of remote sensing data can improve the accuracy of air pollution monitoring results. This is a combination that fills data gaps in areas poorly covered by monitoring stations, and hence, it can offer comprehensive air quality information.

It has been explored to leverage rich spatial information provided by remote sensing data, but the question of how best remotely sensed inputs can be used in conjunction with existing air pollution prediction models remains a challenge. Recent research has recently examined the fusion of remote sensing data with more sophisticated machine learning algorithms, presenting a richer understanding of predicting air pollution. Take MF-net [15] as an example, it employs 3-D remote sensing data to predict pollutants. STI-GCN [16] introduces high-resolution spatial features from remote sensing images for PM2.5 concentrations. SNDNN [14] is a PM2.5 prediction model that uses the CNNs to process MODIS data. Such models log a few victories combined with remotely sensed info and "gut feeling" data from earth-entering surveys.

In monitoring pollutant concentrations, researchers have used remote sensing technology to obtain extensive information on pollutant dispersion. This data, with its wide coverage, contains valuable information about atmospheric pollutant dispersion, compensating for the incompleteness of ground-based pollution monitoring data. However, further research is needed to effectively integrate cross-modal remote sensing data into existing air pollution prediction models.

B. GNNs for Spatial–Temporal Modeling

GNNs have attracted increasing interest and have been applied in many applications due to their superior performance in dealing with complex network-structured data. GNNs naturally model the relations between spatial locations and their temporal evolutions, making them particularly suitable for air pollution prediction tasks with spatiotemporal dependencies [32]. GNNs model graph structures, capturing the spatial relationships between different monitoring points, after which node and edge attributes are used for feature extraction and prediction. Dynamic GNNs [33] extend GNNs by dynamically updating graph structures over time to capture temporal changes and dynamic spatial correlations in data. For instance, GCNs are widely used in traffic prediction [34], [35], weather forecasting [36], and environmental monitoring [37] with good performance enhancement. GCNs extract spatial features and perform convolutions over graph structures, while TCNs can model long-term temporal dependencies because of their dilated convolution.

Indeed, there have been some preliminary attempts to apply GNNs in modeling spatiotemporal features from air pollution data [25], [26], [38]. For instance, researchers could utilize a static GNN to model the spatial distribution of air pollution and a dynamic GNN to capture the temporal dynamics of air pollution. While these methods have improved prediction performance to some extent, they still have some deficiencies. For example, they do not capture spatiotemporal dependencies effectively and require further validation with big data.

In summary, current methods could be improved in their ability to capture spatiotemporal dependencies and optimize the effectiveness of large-scale data. The CMDNet model addresses this by using dynamic GNNs to capture changing spatiotemporal dependencies while integrating remote sensing data to better represent the complex spatiotemporal dynamics of air pollution. This integration has the potential to increase both prediction accuracy and coverage.

III. Preliminaries

A. Problem Formulation

Air pollution prediction is a typical spatiotemporal forecasting problem. In our study, two cross-modal data sources—monitoring sequences and remote sensing images—are jointly utilized to predict future air pollution levels. The historical air pollution sequence data over a time period t is denoted as $X = \{X_1, X_2, \ldots, X_t\}$. Similarly, the remote sensing image data over the same period is denoted as $\mathrm{RS} = \{\mathrm{RS}_1, \mathrm{RS}_2, \ldots, \mathrm{RS}_t\}$. The objective of the prediction model is to learn the spatiotemporal features of the cross-modal data and establish a mapping relationship $\hat{X}_{t+1} = \mathbf{f}(X, \mathrm{RS})$ to predict the air pollution level at the next time step.

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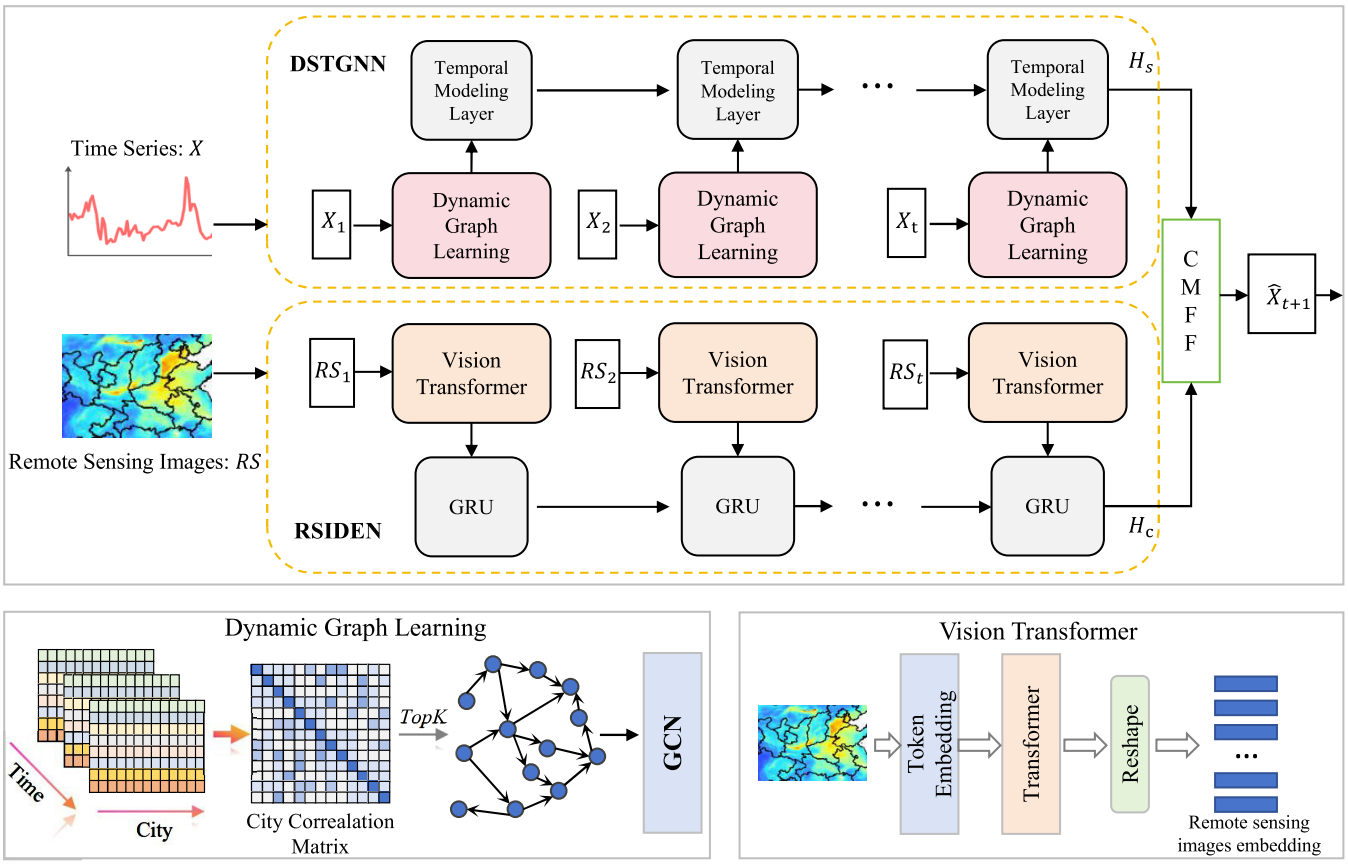


Fig. 3. Framework diagram of the CMDNet, where the input data consists of air pollution monitoring sequence data and remote sensing image data. Dynamic graph learning captures the dynamic correlations between air pollution sequences, while the remote sensing image modeling module learns the semantic features of remote sensing images related to air pollution.

B. Primary Symbols and Definitions

The main symbols used in the model and their definitions are presented below.

1) *X:* Air pollution data in the form of time series, obtained from ground monitoring stations.
2) *RS:* Remote sensing images, collected via remote sensing satellite.
3) H_s*:* The representations learned from time series X by the model.
4) H_e*:* The representations learned from remote sensing images RS by the model.
5) *t:* The time step t for air pollution prediction.
6) *G:* The dynamic graph represents the air pollution relationships between different regions.
7) *A:* The adjacency matrix of the dynamic graph learned by the model, where $\mathbf{A}_{ij}$ represents the relationship between node i and node j.

IV. Methodology

In this section, we present the overall structure of the CMDNet model. As shown in Fig. 3, our approach integrates spatiotemporal dependencies from ground-based air pollution data with spatial features extracted from remote sensing images. The model is composed of three main components: the DSTGNN branch, the RSIDEN branch, and the cross-modal feature fusion (CMFF) module. The DSTGNN captures the dynamic spatiotemporal dependencies of air pollution data, while the RSIDEN extracts semantic information from remote sensing images. The CMFF then fuses these features to enhance the model's predictive performance. This section details the architecture and operations of each component, outlining the processes for dynamic graph construction, feature extraction, and cross-modal fusion. Fig. 4 illustrates the overall flow of the CMDNet model, highlighting the relationships between its different components.

A. Dynamic Spatiotemporal GNN

A key element of our model is the DSTGNN branch, which is intended to gradually learn the dependencies of air pollutants across various cities. The dynamic graph learning layer and temporal modeling layer make up the module's two primary layers. By combining the two parts, the model can capture the temporal as well as the spatial characteristics of air pollution data, leading to more precise air quality predictions.

1) Dynamic Graph Learning: The dynamic graph learning layer constructs a series of dynamic graphs that represent the air pollution correlations of different cities at different time periods. Each graph is composed of nodes and edges, where nodes represent cities and edges represent correlations between these cities. For each time step t, the air pollution correlation graph can be represented as $G_t = (V, E_t)$, where V is the set

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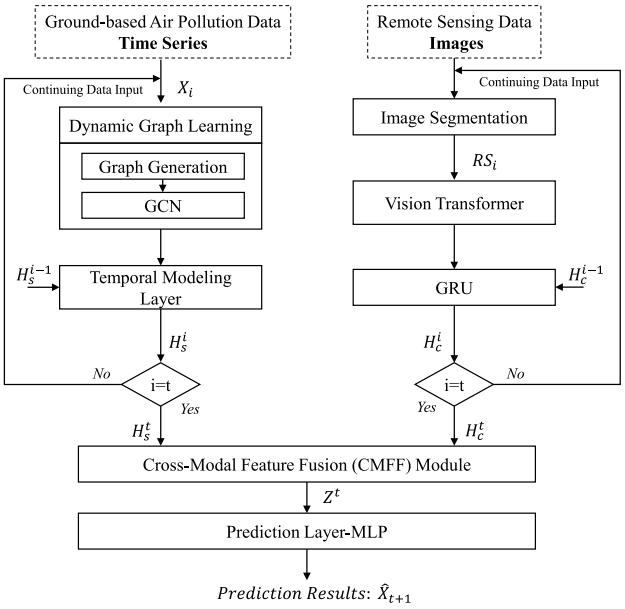


Fig. 4. Overall flow of the CMDNet model.

of nodes representing different cities and E_t is the set of edges representing the correlation relationships. The relationships between nodes in the graph are learned based on air pollution data and geographical barriers.

Based on historical air pollution data, the dynamic graph learning layer models the evolving correlations of air pollution between geographical regions. In the process of constructing the dynamic graph, each node represents a geographical region, and the edges between nodes signify the dynamic relationships in air pollution. $\mathbf{A}_{ij}^{(t)}$ represents the relationship strength between regions i and j at time step t. This relationship is determined using the historical air pollution feature vectors $\mathbf{x}_i^{(t-1)}$ and $\mathbf{x}_j^{(t-1)}$ from the previous time step $t-1$

$$\mathbf{A}_{ij}^{(t)} = \mathbf{similarity}\left(\mathbf{x}_i^{(t-1)}, \mathbf{x}_j^{(t-1)}\right) \tag{1}$$

where $\mathbf{A}^{(t)}$ represents the spatial relationships between geographical regions at time step t. These relationships are computed using the feature vectors $\mathbf{x}_i^{(t-1)}$ and $\mathbf{x}_j^{(t-1)}$, which represent the air pollution data of regions i and j at the previous time step $t-1$. $\mathbf{similarity}(\cdot)$ represents the cosine similarity degree computation function to obtain the correlation or similarity between these feature vectors. To ensure that the adjacency matrix $\mathbf{A}^{(t)}$ retains only the relationships that are important for air pollution prediction, the hyperparameter k is used to control the sparsity of the matrix. Specifically, k represents the top k largest similarity scores in each row. Then, the node pairs corresponding to these similarity scores are selected, and the edges between them are retained in the graph structure.

The objective is to optimize $\mathbf{A}^{(t)}$ to minimize the prediction error of air pollution levels across geographical regions during training. This optimization process involves iterative updates to $\mathbf{A}^{(t)}$ using gradient-based optimization methods, aligning predicted air pollution dynamics closely with observed data. To incorporate geographical barriers, we introduce a geographical information mask matrix M. This mask matrix M has elements m_{ij} that are set to 0 if there is a significant geographical barrier (e.g., a mountain) between cities i and j, and 1 otherwise. The final edge weight matrix is obtained by element-wise multiplication of the learned weight matrix $\mathbf{A}^{(t)}$ and the geographical mask matrix M

$$\hat{A}_t = A_t \odot M. \tag{2}$$

The GNN layer applies graph convolution operations to the dynamic graphs created in the previous layer. The purpose of this layer is to learn the embedding representation of urban nodes from the spatial correlation of air pollution between different cities through graph information aggregation. The GNN layer performs graph convolutions to aggregate information from neighboring nodes. The graph convolution operation can be defined as

$$h_i^{(l+1)} = \sigma\left(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}h_i^{(l)}W^l\right) \tag{3}$$

where $h_i^{(l)}$ is the feature vector of node i at layer l, $\hat{D}$ represents the degree matrix of the adjacency matrix $\hat{A}$, $W^{(l)}$ and $b^{(l)}$ are learnable parameters, and σ denotes the activation function. By performing graph convolution operation, the GNN layer extracts hierarchical spatial features, capturing the complex dependencies between different cities.

2) Temporal Modeling Layer: The temporal modeling layer employs gated recurrent units (GRUs) [26] to capture the temporal dependencies in the sequential air pollution data. The GRU layer processes the sequence of spatial features extracted by the GNN layer, capturing the temporal dynamics and dependencies in the air pollution data across different cities.

By integrating the dynamic graph learning layer and GRU layer, the DSTGNN branch effectively captures the spatiotemporal dependencies in air pollution data across multiple cities. This integration allows the model to learn how air pollution levels in different cities change over time and how these changes are interrelated spatially.

The DSTGNN branch leverages the strengths of dynamic graph construction, GNNs, and recurrent neural networks (RNNs) to provide a robust framework for spatiotemporal modeling of air pollution. By capturing both spatial and temporal dependencies, the DSTGNN branch enhances the predictive power and accuracy of the overall air pollution prediction model. This approach addresses the limitations of traditional models by offering improved spatial coverage and a deeper understanding of the dynamic relationships between air pollutants across different cities.

B. Remote Sensing Image Dynamic Encoding Network

The RSIDEN branch is designed to encode remote sensing images to extract informative features that aid in air pollution prediction. This section details how remote sensing images are processed and encoded to provide temporal and spatial information that complements the DSTGNN branch.

Remote sensing images, containing daily PM2.5 satellite data, are segmented based on the geographical coordinates of the cities under study. This segmentation ensures that the features extracted are directly relevant to the air pollution

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levels in the respective cities. Each city's surrounding area is identified using its latitude and longitude. Remote sensing images are segmented into patches corresponding to these specific areas. For a city located at coordinates $(\text{lat}_c, \text{lon}_c)$, the image I_t for day t is divided into patches $I_t^{(c)}$ where c denotes the city index

$$I_t^{(c)} = \text{Segment}(I_t, \text{lat}_c, \text{lon}_c). \tag{4}$$

This segmentation ensures that the segmented images capture relevant environmental characteristics that impact air quality. Remote sensing images are acquired daily for each segmented area, providing a consistent dataset that reflects daily variations. Each city c has a sequence of images $\{I_t^{(c)}\}_{t=1}^T$, where T is the total number of days. These remote sensing image sequences help in capturing the dynamic nature of air pollution and its influencing factors

$$\{I_t^{(c)}\}_{t=1}^T = \{\text{daily remote sensing images for city } c\}. \tag{5}$$

Vision Transformer (ViT) Encoding: To extract high-dimensional features from the segmented remote sensing images, we utilize a ViT [39] encoding layer. The ViT encodes these images to capture complex spatial patterns relevant to air pollution. Each segmented image $I_t^{(c)}$ is divided into fixed-size patches. Suppose each image is divided into N patches, each patch is flattened and then linearly embedded into a sequence of vectors. Let $\mathbf{x}_p^i$ represent the ith patch of an image, then the embedding process can be formulated as

$$\mathbf{z}_0 = \left[\mathbf{x}_p^1; \mathbf{x}_p^2; \ldots; \mathbf{x}_p^N\right]\mathbf{E} + \mathbf{E}_{\text{pos}} \tag{6}$$

where $\mathbf{E}$ is the embedding matrix and $\mathbf{E}_{\text{pos}}$ is the positional encoding matrix. The embedded patches are processed through the ViT encoder, which consists of multihead self-attention (MSA) and feed-forward neural networks. The core computational processes of the ViT encoder are as follows:

$$\mathbf{z}'_{l+1} = \text{MSA}(\text{LN}(\mathbf{z}_l)) + \mathbf{z}_l \tag{7}$$

$$\mathbf{z}_{l+1} = \text{MLP}(\text{LN}(\mathbf{z}'_{l+1})) + \mathbf{z}'_{l+1} \tag{8}$$

where $\mathbf{z}_l$ represents the output from the lth layer, LN is layer normalization, and MLP is a multilayer perceptron.

The output of the ViT encoding layer provides a sequence of high-dimensional vectors for each day's remote sensing images. By processing a sequence of daily images, the model can learn temporal patterns and trends in the spatial data. For city c, the sequence of encoded vectors is represented as

$$\mathbf{Z}^{(c)} = \left\{\mathbf{z}_t^{(c)}\right\}_{t=1}^T \tag{9}$$

where $\mathbf{z}_t^{(c)}$ represents the encoded feature vector for day t. In addition to spatial feature extraction, it is important to learn the temporal dynamics of remote sensing data. This is achieved by processing the sequence of encoded vectors through an RNN layer such as a GRU layer. The GRU processes the sequence of encoded vectors $\mathbf{Z}^{(c)}$ to capture temporal dependencies. The hidden state of the GRU at time step t, denoted as $\mathbf{h}_t^{(c)}$, is updated as follows:

$$\mathbf{h}_t^{(c)} = \text{GRU}\left(\mathbf{z}_t^{(c)}, \mathbf{h}_{t-1}^{(c)}\right) \tag{10}$$

where $\mathbf{h}_{t-1}^{(c)}$ denotes the hidden state from the previous time step. This enables the model to learn the temporal dynamics of the remote sensing data, capturing how air pollution evolves over time.

C. Cross-Modal Feature Fusion

The CMFF module plays a pivotal role in integrating spatiotemporal features extracted from the DSTGNN branch with dynamic feature representations obtained from the RSIDEN branch. This module is designed to effectively leverage both ground-based monitoring data and satellite remote sensing images. In the feature alignment phase, the module aligns feature dimensions and normalizes feature scales across DSTGNN and RSIDEN outputs to ensure compatibility. To ensure compatibility between the features from DSTGNN and RSIDEN, a rigorous alignment process is employed. This involves temporal synchronization and feature normalization. Temporal alignment ensures that corresponding time steps across different data sources are synchronized, crucial for maintaining temporal coherence in the fused features. Normalization standardizes feature scales, facilitating consistent representation across diverse data modalities. We perform a two-step alignment process to ensure compatibility between the DSTGNN and RSIDEN outputs. Spatial alignment ensures that the spatiotemporal sequence data corresponds to the correct spatial region in the remote sensing images. Temporal alignment addresses the difference in sampling frequencies between the two modalities. The air pollution spatiotemporal sequence data has a shorter sampling interval, so the remote sensing data is replicated along the time axis to align the two modalities. This process obtains the aligned features

$$\mathbf{H}_{\text{aligned}} = \text{Attention}(\text{Norm}(\mathbf{H_s}), \text{Norm}(\mathbf{H_c})). \tag{11}$$

The features from the two branches, DSTGNN and RSIDEN, are combined through a cross-modal fusion mechanism to retain information related to air pollution from both types of data. The module explores various fusion techniques, including concatenation and attention mechanisms. Concatenation preserves the individual characteristics of each feature set while combining them into a unified representation. In contrast, attention mechanisms allow for selective integration of features based on their relative importance, dynamically adjusting the fusion process to emphasize critical information. Then, we employ a weighted summation to integrate the remote sensing spatiotemporal features $\mathbf{H}_{\text{aligned}}$ and spatiotemporal features $\mathbf{H}_s$. The coefficient $\boldsymbol{\alpha}$ determines how much each modality contributes to the final integrated feature

$$\mathbf{Z} = \boldsymbol{\alpha} \cdot \mathbf{H}_{\text{aligned}} + (1 - \boldsymbol{\alpha}) \cdot \mathbf{H}_s \tag{12}$$

where $\mathbf{Z}$ represents the fused feature representation that integrates comprehensive spatiotemporal dependencies and remote sensing semantic information. This enhanced representation enables the model to predict the concentration of air pollutants accurately.

By integrating the features of two modal data, the CMFF module provides a comprehensive representation of air pollution dynamics. This enhanced feature representation captures

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both the detailed spatiotemporal dependencies modeled by DSTGNN and the semantic information derived from RSIDEN's remote sensing images. The module's output serves as input to the prediction layer, which utilizes regression-based loss functions to optimize model parameters to accurately predict the concentration of air pollutants

$$\hat{y} = f_{\text{pred}}(\mathbf{Z}) \tag{13}$$

where $\hat{y}$ denotes the predicted values of air pollutants and f_{pred} represents the prediction model: CMDNet. We use the mse loss as the loss function for the CMDNet model, and the formula for this loss is as follows:

$$\text{mse} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2 \tag{14}$$

where N represents the number of nodes, that is, the number of cities in our prediction model, $\hat{y}_i$ and y_i are the predicted and actual values of air pollutant concentrations in city i, respectively.

In summary, the CMFF module integrates and utilizes information related to air pollution from data across different modalities. This method enhances the model's ability to capture the complex spatiotemporal factors affecting air quality and their interactions, thereby improving prediction accuracy.

V. Experiments

To validate the effectiveness of the CMDNet model for air pollution prediction, We conduct comprehensive experiments, including comparisons with baseline models, detailed analysis of prediction results, and ablation studies to assess the contribution of each component.

A. Datasets and Setup

Our dataset comprises ground station monitoring data and remote sensing images, encompassing three types of monitoring data: PM2.5, NO_2, and SO_2. The PM2.5 data is sourced from the KnowAir dataset [25], while the NO_2 and SO_2 data are obtained from monitoring data of an open-air quality data site.[1] The sampling interval for PM2.5 data is every three hours, whereas the NO_2 and SO_2 data are recorded daily. The time period for PM2.5 data covers the entire year of 2016, while NO_2 and SO_2 data span the entire year of 2022. The corresponding remote sensing image data aligns with the temporal periods of the respective pollutant data.

The remote sensing image data is derived from the China High-resolution Air Pollution (CHAP) dataset,[2] which is a high-quality, high-resolution dataset for environmental air pollutants in China. We obtained ChinaHighPM2.5 [40], ChinaHighNO_2 [41], and ChinaHighSO_2 [41], which provide daily air pollution monitoring data with a spatial resolution of 1 km.

[1]https://www.aqistudy.cn/

[2]https://zenodo.org/communities/chap/records

1) Cross-Modality Data Processing: Based on the two data sources mentioned above, we constructed a cross-modality air pollution prediction dataset. This dataset consists of three subsets: PM2.5, NO_2, and SO_2. Each subset contains sequence data and remote sensing image data on air pollution for specific time periods. To ensure temporal consistency between the two types of data, we aligned them during the data processing. Specifically, we processed the remote sensing data to match the time periods and spatial locations of the sequence data. We applied normalization methods to standardize the data across different modalities. Then, we split the processed dataset into training, validation, and test sets in an 8:1:1 ratio.

2) Experimental Settings: All experiments were conducted on a server equipped with a single NVIDIA 3090 GPU (24 GB VRAM), and the model was built based on the PyTorch framework. In the DSTGNN module, the number of GCN layers was set to 2, and the initial hidden state h_0 of the GRU layers in the model was initialized with a zero tensor. The model training was performed on three datasets for a total of 150 epochs, with a learning rate of 0.0001 and a batch size of 64. All input data was normalized before training, with each feature being standardized based on its mean and standard deviation in the training set. The input data step size was 1 step (i.e., 3 h), and the prediction output consisted of 24 steps (i.e., 72 h). The experimental results were repeated ten times, and the average value was taken.

3) Evaluation Metric: To evaluate the performance of our proposed air pollution prediction model, we adopted five evaluation metrics commonly used in air pollution forecasting, as referenced from PM25-GNN [25]. These metrics include MAE and RMSE, which are commonly used in time-series forecasting, as well as CSI, FAR, and POD, which are used in weather forecasting. CSI represents the proportion of correct positive predictions among all predicted or actual occurrences. FAR represents the proportion of false positives among all positive predictions, reflecting the model's false alarm rate (FAR). POD represents the ratio of true positives among all correctly predicted results by the model. Below are the formulas for calculating these evaluation metrics

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{15}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{16}$$

$$\text{CSI} = \frac{n_{\text{hit}}}{n_{\text{hit}} + n_{\text{miss}} + n_{\text{falsealarm}}} \tag{17}$$

$$\text{FAR} = \frac{n_{\text{falsealarm}}}{n_{\text{hit}} + n_{\text{falsealarm}}} \tag{18}$$

$$\text{POD} = \frac{n_{\text{hit}}}{n_{\text{hit}} + n_{\text{miss}}} \tag{19}$$

where y_i and $\hat{y}_i$ represent the actual value of the air pollutant and the output value of the CMDNet model, respectively. n denotes the total number of observations. To evaluate the CSI, POD, and FAR metrics, we set thresholds for various types of air pollutants to categorize air quality into severe pollution or normal levels. n_{hit} is the count of accurate severe pollution

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TABLE I
PERFORMANCE COMPARISON RESULTS BETWEEN CMDNET AND BASELINE MODELS FOR AIR POLLUTION PREDICTION, WITH BOLD NUMBERS INDICATING THE BEST RESULTS

Method		ARIMA	MLP	GRU	LSTM	GC-LSTM	GWNet	STSGCN	PM25GNN	GAT-GRU	Airformer	DDGCRN	TGCRN	CMDNet
PM2.5	MAE	45.74	42.23±0.3	38.11±1.1	38.04±0.6	37.99±0.9	36.02±0.4	36.80±0.7	36.69±0.2	34.56±0.5	32.75±0.1	34.35±0.8	33.10±0.3	**32.09±0.2**
	RMSE	59.14	51.25±0.9	47.20±0.2	47.19±1.0	46.30±0.7	43.67±0.3	44.10±0.6	44.90±0.8	42.79±0.4	41.96±0.2	42.60±0.5	42.28±0.9	**41.22±0.1**
	CSI	50.10	51.79±0.4	60.34±0.9	59.66±0.3	59.20±0.6	60.23±0.8	59.84±0.2	60.02±0.5	61.71±0.1	62.80±0.7	61.58±0.4	62.27±0.9	**63.21±0.3**
	FAR	37.98	36.10±0.8	33.74±0.3	32.01±0.5	32.77±0.2	31.33±0.9	32.04±0.4	31.98±0.7	28.55±0.1	29.14±0.6	28.71±0.8	29.23±0.3	**28.17±0.2**
	POD	69.23	73.20±0.7	81.20±0.2	80.01±0.9	80.96±0.5	82.17±0.3	81.83±0.8	83.11±0.1	81.95±0.6	83.54±0.4	82.33±0.7	83.92±0.2	**84.60±0.5**
NO_2	MAE	50.12	47.05±0.2	44.3±0.7	43.89±0.4	43.21±0.8	41.78±0.3	42.56±0.6	42.09±0.1	40.23±0.5	38.92±0.9	39.26±0.2	39.55±0.6	**38.45±0.4**
	RMSE	64.20	59.78±0.8	54.67±0.5	54.95±0.2	54.38±0.7	52.14±0.9	52.89±0.4	53.10±0.6	51.34±0.7	**49.87±0.5**	50.96±0.8	50.20±0.3	49.91±0.7
	CSI	48.56	51.32±0.9	57.89±0.4	57.34±0.7	57.57±0.2	58.78±0.5	58.23±0.8	58.67±0.3	59.56±0.5	60.13±0.6	59.81±0.9	60.30±0.4	**60.98±0.2**
	FAR	40.01	37.22±0.8	35.16±0.3	33.98±0.6	34.56±0.9	33.19±0.2	34.09±0.7	33.47±0.4	**30.67±0.1**	31.45±0.5	31.88±0.7	30.93±0.2	30.72±0.8
	POD	66.45	71.34±0.4	78.90±0.9	76.43±0.2	76.78±0.6	77.12±0.5	76.93±0.8	78.34±0.3	76.45±0.2	78.30±0.7	77.46±0.4	78.09±0.9	**79.45±0.2**
SO_2	MAE	35.67	33.49±0.8	31.23±0.5	30.89±0.2	30.34±0.7	29.01±0.4	29.78±0.9	29.34±0.6	28.12±0.1	26.89±0.3	27.14±0.8	26.66±0.5	**26.01±0.2**
	RMSE	48.76	45.28±0.4	42.12±0.7	42.45±0.2	41.89±0.5	40.12±0.8	40.89±0.3	41.12±0.6	39.45±0.1	38.39±0.9	38.12±0.4	38.83±0.7	**37.45±0.2**
	CSI	52.89	55.67±0.7	62.34±0.4	61.78±0.9	62.01±0.3	63.23±0.6	62.78±0.5	63.12±0.8	64.01±0.3	65.24±0.2	64.70±0.7	65.09±0.4	**66.12±0.3**
	FAR	36.01	33.78±0.5	31.45±0.2	30.12±0.7	30.78±0.4	29.56±0.9	30.34±0.3	29.89±0.6	27.45±0.4	28.34±0.8	27.90±0.5	28.05±0.2	**27.01±0.4**
	POD	70.12	74.45±0.3	82.01±0.8	79.45±0.5	80.12±0.2	80.67±0.7	80.45±0.4	81.34±0.9	80.23±0.1	81.89±0.6	81.27±0.3	81.53±0.8	**82.34±0.2**

predictions, n_{miss} indicates the missed severe pollution events, and $n_{\text{falsealarm}}$ counts the false alarms where severe pollution was predicted but did not occur. These evaluation metrics collectively provide a comprehensive assessment of the model's predictive performance to accurately forecast pollutant concentrations.

B. Baselines

We utilized the following methods as baselines for comparison.

ARIMA [42]: As a classical time forecasting model, ARIMA combines both autoregressive and moving average methods.

MLP [43]: MLP is a basic neural network model that takes pollution data as input to output future air pollution values.

GRU [26]: As a classic variant of the RNN, GRU is widely employed in time prediction and commonly used in crime occurrence prediction research.

LSTM [26]: LSTM is also a classic variant of the RNN, used for time-series prediction.

GC-LSTM [38]: GC-LSTM predicts air pollutant concentrations by combining the GCN and LSTM.

GWNet [44]: Graph WaveNet (GWNet) is a predictive model used for spatiotemporal graph modeling. It accurately captures spatial correlations in spatiotemporal data by learning an adaptive dependency matrix that describes the relationships between nodes.

STSGCN [45]: STSGCN is designed to capture local complex temporal dependencies through a spatiotemporal synchronous modeling mechanism.

PM2.5-GNN [25]: PM2.5-GNN is a graph network-based air pollution prediction model that incorporates domain knowledge of air pollution to capture long-term dependencies.

GAT-GRU [26]: GAT-GRU combines graph attention networks and RNNs to achieve air pollution prediction.

DDGCRN [46]: The DDGCRN model adaptively learns evolving graph structures from time-varying signals and effectively captures spatiotemporal patterns.

TGCRN [47]: TGCRN proposes a time-aware time-series prediction method by calculating the interaction relationships between time-series features, which we apply to the air pollution prediction task.

Airformer [24]: Airformer employs two novel self-supervised mechanisms to capture the spatiotemporal features.

C. Forecast Results and Analysis

This section presents a comparative analysis of our proposed CMDNet model against 12 baseline models on three datasets: PM2.5, NO_2, and SO_2. As shown in Table I, the CMDNet performs exceptionally well across all evaluation metrics on the PM2.5 dataset. Specifically, the MAE of CMDNet is 32.09, which surpasses all other baseline models. Compared to the SOTA model Airformer (32.75), it has decreased by approximately 2%. In terms of RMSE, the CMDNet achieves the best performance with an RMSE of 41.22, which is 1.8% lower than Airformer (41.96), indicating higher predictive accuracy. Additionally, the CMDNet records a CSI of 63.21, significantly outperforming other models, demonstrating its superior capability in capturing high pollution events. For the FAR metric, the CMDNet achieves the lowest rate of 28.17, indicating the most accurate predictions with the least false alarms. Lastly, the CMDNet achieves the highest POD of 84.60, showing its strong ability to detect pollution events.

On the NO_2 dataset, the CMDNet also performs outstandingly on most metrics. Specifically, CMDNet's MAE is 38.45, which is the best result, reducing by 1.2 % compared to Airformer (38.92). Despite the RMSE of the CMDNet being 49.91, nearly identical to Airformer (49.87), it maintains a low error. The CMDNet achieves a CSI of 60.98, outperforming all baseline models and demonstrating the best capability in capturing high-pollution events. For FAR, the CMDNet records 30.72, second only to GAT-GRU (30.67), indicating a low FAR. Finally, the CMDNet achieves the highest POD of 79.45, showing its superior ability to detect pollution events.

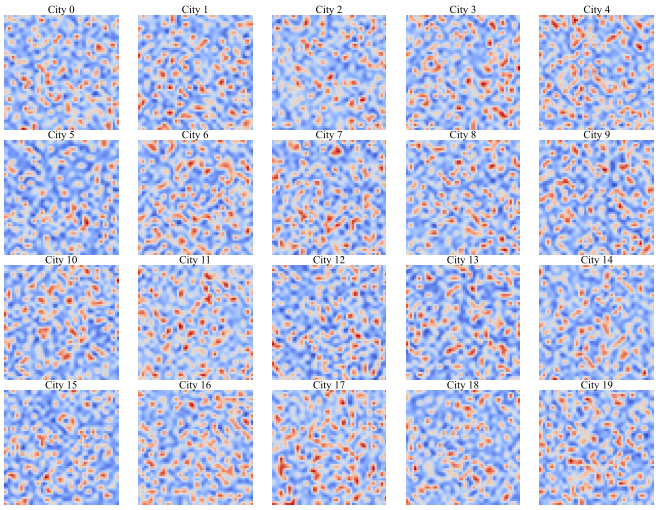
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On the SO_2 dataset, the CMDNet performs excellently. Specifically, similar to the performance on the first two datasets, the CMDNet achieved an MAE of 26.01 on SO_2, which is a 2.4% reduction compared to TGCRN (26.66). In terms of RMSE, the CMDNet records the best performance with an RMSE of 37.45, which is 1.8% lower than DDGCRN (38.12). Additionally, the CMDNet achieves a CSI of 66.12, significantly outperforming all baseline models, demonstrating its superior capability in capturing high pollution events. For FAR, the CMDNet records the lowest rate of 27.01, indicating the most accurate predictions with the least false alarms. Lastly, the CMDNet achieves the highest POD of 82.34, showing its strong ability to detect pollution events.

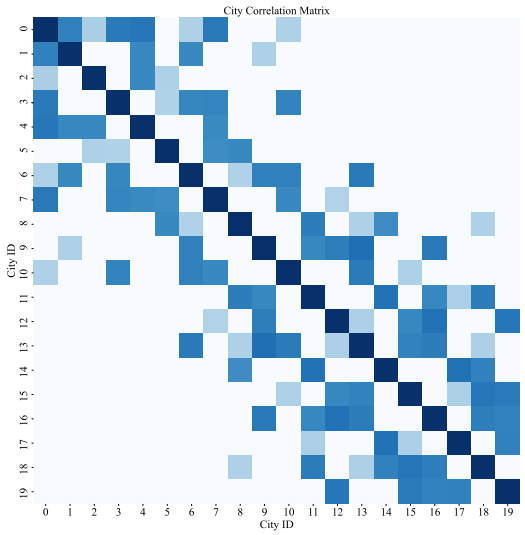
In our study, GRU is employed to capture the temporal dynamics of both air pollution sequences and remote sensing representations. Compared to other sequence modeling approaches, GRU demonstrates superior predictive performance on the PM2.5 and SO_2 datasets, while its performance on the NO_2 dataset is slightly inferior to the Transformer. However, considering the overall results across multiple datasets, GRU provides better generalization and stability. The gating mechanism in GRU effectively models long-term dependencies while maintaining computational efficiency, making it well-suited for air pollution prediction. These advantages justify our choice of GRU as the temporal modeling layer in the CMDNet model.

Fig. 5(a) visualizes the embedding representations learned by the RSIDEN branch in the CMDNet model. We selected PM2.5 remote sensing data from 20 cities to intuitively illustrate the spatial distribution characteristics of PM2.5 across different urban areas. These visualizations reveal the spatial distribution patterns of air pollution in urban surroundings and can be integrated with ground-based monitoring data to enhance the accuracy of pollution prediction. Furthermore, Fig. 5(b) presents the correlation matrix learned through the dynamic graph learning layer in the DSTGNN branch for these 20 cities. The color in the figure ranges from white to blue, indicating correlation strength from weak to strong. It can be observed that the correlation matrix aligns well with the geographic distribution of the cities—cities indexed from 0 to 11 are primarily located around Beijing, while those with higher indices are situated in southern China. Overall, cities in close geographical proximity exhibit stronger correlations in air pollution patterns, further validating the model's effectiveness in capturing spatial dependencies.

Comparative experiments demonstrate that the CMDNet model has outperformed existing baseline models in air pollution prediction performance, particularly in terms of predictive accuracy (MAE and RMSE), the capability to capture high-pollution events (CSI and POD), and reducing FARs. This indicates that the combination of the DSTGNN with remote sensing data provides significant advantages in air pollution prediction tasks, offering more accurate and reliable predictions. These results validate the effectiveness and robustness of our proposed method.



(a)



(b)

Fig. 5. Subfigures (a) and (b) are both based on the PM2.5 data of 20 cities, presenting the visualization of remote sensing image embeddings and the correlation matrix between cities, respectively. (a) Visualization of remote sensing image embeddings for some cities learned by RSIDEN. (b) Adjacency matrix of city correlations for some cities obtained through dynamic graph learning.

D. Hyperparameter Analysis

In this experiment, we analyzed the sparsity hyperparameter k in the dynamic graph learning layer, selecting $k = [2, 3, 4, 5, 6]$ as candidate values, and tested the model on the SO_2, NO_2, and PM2.5 datasets using MAE and RMSE as performance evaluation metrics. Fig. 6 presents the experimental results. It is important to note that smaller k values correspond to sparser graph structures. This means that with smaller k values, fewer adjacent nodes are retained in the graph, and the model only preserves the most critical local connections. In the PM2.5 dataset, the model achieved its best performance when $k = 3$, with the minimum MAE of 32.09 and RMSE of 41.22, indicating that a moderately sparse graph structure can effectively capture key features in the PM2.5 data. In the SO_2 dataset, the minimum MAE and

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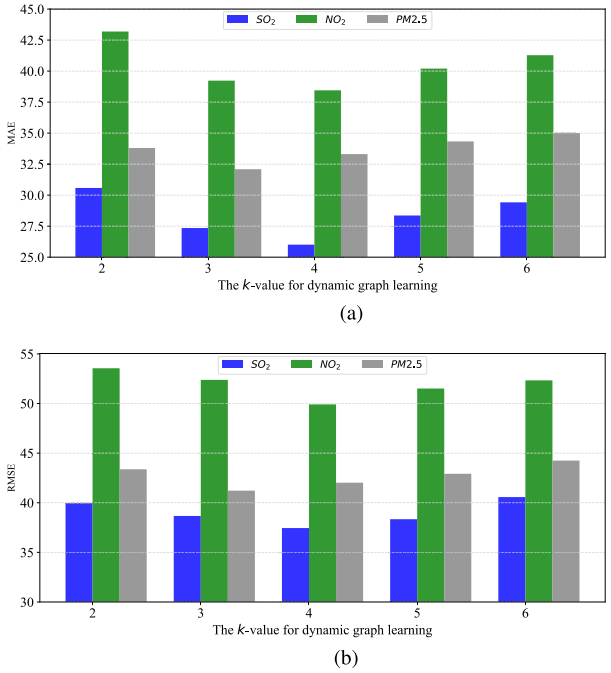


Fig. 6. Effect of hyperparameters on MAE and RMSE across three datasets. (a) Results of the hyperparameter study experiment (MAE). (b) Results of the hyperparameter study experiment (RMSE).

RMSE values of 26.01 and 37.45 were achieved when $k = 4$, suggesting that increasing the graph structure density appropriately helps capture more complex spatiotemporal dependencies in the SO_2 data. Similarly, in the NO_2 dataset, the best results were achieved at $k = 4$, with MAE and RMSE values of 38.45 and 49.91, respectively. This indicates that a moderately dense graph structure is more suitable for learning features in the NO_2 data.

E. Computational Time Study

To further evaluate the practicality of the CMDNet, we compare its computational efficiency with baseline models (PM25GNN and Airformer) on the NO_2 dataset. All experiments are conducted on a 3090 GPU server. The results are summarized as follows.

Training Time: CMDNet requires 24 s per epoch, which is slightly longer than PM25GNN (17 s) but comparable to Airformer (22 s). This indicates that the CMDNet maintains a reasonable training efficiency while achieving superior performance.

Inference Time: For single-sample inference, the CMDNet takes 3.5 s, which is marginally higher than PM25GNN (2 s) but remains competitive with Airformer (3 s). This demonstrates that the CMDNet can deliver accurate predictions without significantly compromising inference speed.

These results suggest that the performance improvements of the CMDNet are achieved with a moderate increase in computational cost.

F. Ablation Study

In our ablation study, we designed several model variants to evaluate the impact of different components on the overall performance. The baseline model, CMDNet, includes all components and serves as the reference point for comparison. Each variant systematically removes or alters specific components to assess their contributions to the model's performance.

CMDNet: This model includes all components: DSTGNN, RSIDEN, and CMFF. It serves as the standard against which other variants are compared.

w.o DSTGNN: In this variant, the DSTGNN branch is removed, and only features from the RSIDEN branch and CMFF branch are used. This setup aims to evaluate the importance of capturing dynamic spatiotemporal dependencies from ground-based data.

w.o RSIDEN: Here, the RSIDEN branch is removed, utilizing only the DSTGNN branch and CMFF module. This variant assesses the value of incorporating remote sensing images for spatial feature extraction.

w.o CMFF: This variant directly concatenates features from the DSTGNN branch and RSIDEN branch without using the CMFF, determining the effectiveness of the CMFF mechanism.

w.o Time: In this variant, the temporal modeling layer (GRU) is removed from the DSTGNN branch, retaining only the dynamic graph learning and GNN layers. This setup aims to understand the contribution of temporal modeling in capturing time-dependent air pollution dynamics.

w.o DGL: The dynamic graph learning layer is replaced with a static graph structure in the DSTGNN branch for this variant, evaluating the importance of dynamically learning air pollution correlations over time.

w.o ViT: In this variant, the ViT layer in the RSIDEN branch is replaced with a simpler CNN-based encoder. This change assesses the impact of using advanced feature extraction techniques for remote sensing images.

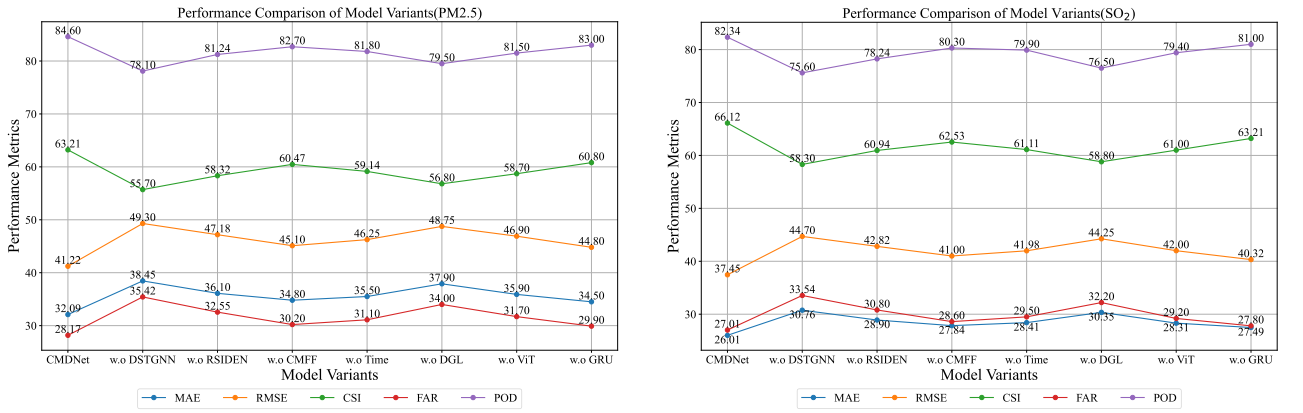
w.o GRU: This variant removes the GRU layer from the RSIDEN branch, processing remote sensing features as static inputs. It aims to determine the role of temporal modeling in the RSIDEN branch for air pollution prediction.

The ablation study provides a comprehensive examination of the contributions made by each component in our CMDNet model for predicting air pollution levels. By systematically removing individual components and evaluating the impact on the model's performance, we gain insights into their importance and functionality.

As illustrated in Fig. 7, our full CMDNet model, which integrates the DSTGNN branch, RSIDEN branch, and CMFF module, demonstrates superior performance. This model achieves an MAE of 32.09, indicating high accuracy in predictions. The RMSE of 41.22 highlights the model's robustness in handling variability in air pollution data. The values of CSI and POD underscore the model's efficacy in accurately identifying pollution events, with CSI at 63.21 and POD at 84.60. Additionally, the FAR is maintained at a low level of 28.17, demonstrating the model's precision in minimizing false positives.

When the DSTGNN component is removed (**w.o DSTGNN**), there is a marked increase in both MAE and RMSE, alongside a decrease in CSI and POD, and an increase in FAR. This degradation in performance underscores the critical role of DSTGNN in capturing the dynamic spatiotemporal

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(a) (b)

Fig. 7. Ablation study results. (a) Performance of different model variants on the PM2.5 dataset, while (b) illustrates the results on the SO_2 dataset. (a) Results on the PM2.5 dataset. (b) Results on the SO_2 dataset.

dependencies essential for precise air pollution predictions. The **w.o DSTGNN** variant indicates that the model only contains the CMFF and RSIDEN modules, relying solely on remote sensing images to predict air pollution. The absence of DSTGNN leads to a significant performance decline, as the model can only utilize the spatial features from the images. Similarly, excluding the RSIDEN branch (**w.o RSIDEN**) results in notable performance deterioration. The increases in MAE and RMSE, along with the declines in CSI and POD, indicate the essential contribution of RSIDEN in extracting semantic features from remote sensing data. The **w.o RSIDEN** variant means the model relies solely on the CMFF and DSTGNN modules, using time-series data for prediction. Although DSTGNN captures spatiotemporal dependencies, the absence of remote sensing features from RSIDEN limits the model's performance. The CMFF module, which is responsible for integrating features from both DSTGNN and RSIDEN, also proves to be indispensable. The removal of the CMFF module (**w.o CMFF**) leads to performance declines, emphasizing the necessity of effective CMFF. The **w.o CMFF** variant removes the CMFF module and directly adds the outputs from RSIDEN and DSTGNN. While this reduces computational complexity, it loses the effective fusion of cross-modal features. The CMFF ensures that the complementary information from spatiotemporal and remote sensing data is harnessed optimally, enhancing the model's predictive capabilities. These three variants clearly highlight the important interactions between modules: DSTGNN is crucial for capturing spatiotemporal dependencies, and its absence significantly reduces performance. RSIDEN provides essential semantic features of pollution distribution, and without it, the model can only rely on node-level sequence data. CMFF is responsible for integrating features from both modalities, and its removal diminishes the model's performance.

Furthermore, the temporal modeling layer using GRUs within the DSTGNN branch (**w.o Time**) has a significant impact on prediction performance. The exclusion of this layer results in increased prediction errors, underscoring the importance of capturing temporal dependencies in the air pollution data to accurately model changes over time. The dynamic graph learning layer within DSTGNN also plays a vital role. Its removal (**w.o DGL**) causes a substantial drop in performance metrics, highlighting the importance of modeling evolving correlations between geographical regions. This layer is crucial for understanding the spatial dynamics

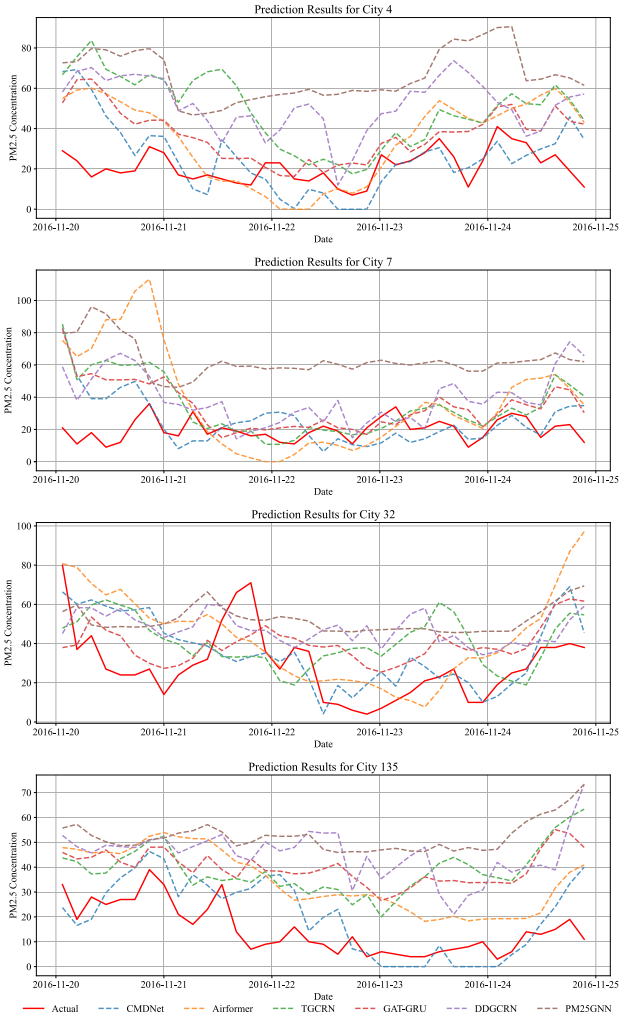


Fig. 8. Comparison of prediction results between the CMDNet and baseline models in different cities. The red solid line represents the ground truth, the blue dashed line represents CMDNet's predictions, and the other dashed lines represent the results of baseline models.

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of air pollution and its propagation across different areas. The ViT used in the RSIDEN branch for extracting complex spatial patterns from remote sensing images is another key component. Without the ViT, the model's ability to process and utilize high-dimensional spatial data from satellite images is significantly compromised, leading to inferior prediction performance. Lastly, the GRU within RSIDEN, responsible for modeling the temporal dynamics of remote sensing data, is essential. Its exclusion (**w.o GRU**) leads to degraded performance, indicating the importance of capturing how environmental features evolve over time to enhance the accuracy of air pollution predictions.

The ablation study clearly demonstrates that each component of the CMDNet model is indispensable. The integration of the DSTGNN branch, RSIDEN branch, and CMFF module collectively contribute to the model's high accuracy and robust performance.

G. Visualization of Prediction Results

Fig. 8 presents a visualization comparing the prediction results with the ground truth. From top to bottom, the four subplots display the prediction results for different cities over the time period from November 20 to November 25, 2016. In the figure, the red solid line represents the actual air pollutant concentration values, while the dashed lines in different colors indicate the predictions of various models. The visualization shows that the CMDNet model, represented by the blue dashed line, produces predictions that are closer to the red solid line compared to other baseline models. The results across different cities consistently demonstrate that the proposed model achieves the best predictive performance.

VI. Conclusion and Future Work

In this article, we proposed a CMDNet model that enhances air pollution prediction using remote sensing images. By integrating DSTGNNs with remote sensing images, the model effectively captures spatiotemporal dependencies and enhances spatial coverage, leading to improved prediction accuracy. Ablation experiments confirm the significant contributions of each component, highlighting the importance of dynamic spatiotemporal graph learning and advanced feature extraction techniques. For the problem of cross-modal spatiotemporal prediction, the CMDNet model proposes a general research paradigm. The model separately models spatiotemporal data of different modalities and integrates spatiotemporal representations across modalities. Therefore, this model can be applied to different fields such as weather prediction, traffic prediction, and environmental prediction.

Future work could focus on incorporating additional data, such as meteorological and real-time traffic information, to further enhance predictive capabilities. Additionally, extending the model's applicability to different regions and pollutant types would help validate its robustness and generalizability.

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